

# ActInf Textbook Group ~ Cohort 2 ~ Meeting 6 (Chapter 3, part 1)

*TextbookGroup/ParrPezzuloFriston2022/Cohort\_2 | Cohort 2 ~ Meeting 6 (Chapter 3, part 1) | 59:38*

Okay, hello everyone.

It's October 7th, 2022.

We're in the sixth meeting for the second cohort of the textbook group.

Today we're going to be discussing chapter three for the first time.

So we have a few questions to explore on three.

But before we do, would anyone like to just say hello or give any overview thoughts on chapter three before we go into more specific questions?

You can raise your hand or just unmute and just go for it.

I'll wait for someone to make a comment on chapter three.

Thank you.

Yep.

Two active inference with the principle of least action and the introduction of Markov blankets.

Okay.

How does that differ from previous approaches or relative to what?

Well, I guess instead of starting with the notion that you have some agent that's creating a representation of its world,

you start with like an axiomatic view from physics that when applied to biological entities, leads to at least the same conclusion as chapter two.

All right.

Awesome.

Thank you.

Anyone can raise their hand or just go for it.

General thoughts on chapter three and or what is the high road to active inference?

Thank you.

Having read chapters two and three.

How does anybody want to remark on figure 1.2?

Thank you.

Okay.

I'll make a short remark.

To me, this figure, I guess, these elements like surprise minimization, Markov blanket, it looks like they are inserted in the pictures like sequentially, but it doesn't seem to me that they actually represent a sequence for understanding the approach. They're just various aspects of either the physics approach or in the lower part of the figure of the Bayesian approach.

If I understood it correctly.

If I understood it correctly.

Because at first I thought it was like a path, you know, starting from the free energy principle, then first to self-organization, then surprise minimization, Markov blanket, predictive processing.

And I don't know if it's, if it is actually a progression of concepts or actually, I don't know.

What do you think?

Yeah.

It's a great question.

Anyone can give a thought.

What do the ordering of the terms mean?

Are these train stops along the way blue or child?

So I just wanted to say that the live stream where Maria really gave a lot of background on predictive processing,

predictive coding, like the difference between these two might be super helpful with regard to like a progression

or just evaluating that lineage in more detail.

Yeah, sure.

I'll add link there.

One option would be that these are historical.

Like this is this chronology of ideas.

Another one would be that they're pedagogical, that this is a learning path.

Another might be formal or technical, like one of them are one of the directions are dependencies or required, but not necessarily the most pedagogical path.

Ali?

I also think that the organization of this particular figure is more like the branch of a tree than something sequential.

So they might not represent any specific preference to go through all of these steps in order to achieve a particular goal.

They're just try it just tries to lump together relevant concepts in two different branches.

So that's what I see it at least.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

That sounds weird.

from 2010 with a lot of the early sort of directions and pointers towards areas of synthesis.

Anyone else want to just add any other thoughts like what is the high road

or how did they feel about chapter three overall?

There's ample technical and specific questions to dive into, but it's really important to understand chapter two and the low road and chapter three being the high road and the way that they're distinguished in terms of the low road beginning from a framework for doing statistics, learning, updating, and so on, but not necessarily an imperative for survival. In contrast, in the high road, the beginning stance is around the imperative for survival, which is kind of a biological way to frame it, or the slightly even more deflationary imperative for repeated measurements, which is more of a quantum or a physics way to be framing the issue in terms of stochastic processes and measurement theory, where if something cannot be measuring itself, it's not going to see itself around for long. And if something is not a thing to something else, it's not being measured over a given timescale within a given variance criterion. It also is going to be, quote, not a thing. So this imperative for persistence and finding manifolds of stability or bounded spaces is an imperative that isn't intrinsically within Bayesian theorem, but active inference is being situated at the intersection of the Bayesian methods and the imperative for persistence and or repeated measurements. Ali?

Yes. One of the, I think, the most, probably the most central concept of chapter three, the, on which all the other material seems to, seems to evolve, is the concept of Markov blanket, but, but in this particular chapter, I think the Markov blanket, I mean, the nature of Markov blanket, has not made quite explicit, so I just wanted to emphasize a critical point about this whole concept, which has caused a lot of, lots of confusions and misunderstandings around it. And, namely, some people think Markov blanket is actually some kind of spatial temporal boundary around the system.

But, as the paper on Bayesian mechanics makes clear, Markov blanket is actually a boundary, a boundary, a boundary on state space, and it doesn't necessarily, or it doesn't always translate into any spatial temporal boundary. In some cases, for example, in the cells or organelles or things like that, we can see

the spatial temporal manifestation of Markov blanket, but that's not always the case. So, yeah, that's one of the points, I think, is crucial to understanding this whole high road approach to active inference.

Markov blanket is a very, very interesting point.

So, they do, in the first box of the chapter and in the first non-introduction section, discuss Markov blankets. So, it's definitely something that we want to follow on. So, let's, we're done for now with a just general discussion around what the high road is.

Good.

Yes, Giuseppe, please.

All right. No, it's just on this topic that it's true that the Markov blanket is explained quite clearly, if succinctly in that part, but then I don't think it throughout the book is actually used or described in details from the computational point of view.

That, I mean, I think it's described, it's defined there, but then to me, since that doesn't appear much in the more computational later parts of the book, that was my impression.

So, the concept is quite clear,

but I'm not sure how it applies then practically, just in laying out the model probably, in laying out the specifics of the model that defines by itself the relevant Markov blanket, probably.

Thank you. Ali?

Yes, one reason for that might be, at least according to my personal communication with Maxwell Ramstad, you see the problem of identifying Markov blankets from data is still an outstanding issue and problem among FEP researchers on which VersusLab is actively working on.

So that's not a resolved problem I guess yet so that's I think one of the reasons they have not alluded to the concept of marker blankets so much throughout the whole throughout the rest of the book the other reason is the recent literature somehow tries to reduce its dependence on this whole concept of markup blanket because of various reasons so yeah these two reasons might be sufficient to explain its lack of appearance or at least explicit appearance in much of the book. Thank you. Great yeah I think this this question will help us dive further into this question. On page 43 they are writing so first off anyone who wanted to describe on a more technical note we can bring it up later or sooner what a markup blanket is because it is one of the core terms. In brief a markup blanket is the set of variables that mediate all statistical interactions between a system and environment and those parentheses cover a lot.

That is not the same thing as the actual mechanical or material interactions between the system and the environment. That is the map territory fallacy or failure to distinguish although one can't also swing too far into the map territory fallacy fallacy area which we've talked about earlier.

This the markup blanket and what it does is a restatement of the classical action perception cycle where an adaptive system and its environment can interact only through actions and observations respectively.

So how is the markup blanket a restatement of the classical action perception cycle?

I feel that this is an important question to understand to contextualize active inference within the history of models like cybernetics control theory ecological psychology and qualitative and quantitative models of perception and action. Kind of goes without saying but it's a really important note and that we see multiple representations of the action perception loop. So figure 3-1 shows the action perception loop

along with the particular partition. That's the one that's separated into internal and external states along with the blanket states which makes internal and external conditionally independent. And then the blanket states

are further decomposed into active states those with outgoing dependencies from the perspective of internal states and sensory states those with internal dependencies. Although note some asterisks in the interpretation for example around which arrows have two directions and which ones don't. So here's figure 3.1.

The Markov blanket is dynamic it's changing in time the states are changing in time and separating, separating, partitioning, internal states here the brain from the environment. And all of the functions are described in terms of rates of change  $\dot{x}$ ,  $\dot{u}$ ,  $\dot{y}$ , and  $\dot{\mu}$  and the dot over the top is like the derivative. So they're all being solved for in terms of how they change through time and they're all being described with these  $f$  functions. So they're all being solved for in terms of the equation. So they're all being solved for the equation. So they're all being solved for the equation. So they're all being solved for the average rate of change over these arguments coming into the deterministic component of the equation. And  $\omega$  which is some sort of random fluctuation parameter. So it's kind of like a signal to noise that's the same. So we're going to see the change of the equation. So we're going to see the change characteristics of a given state corresponding to particular partitioned variables. But we've also seen other representations of the action perception loop. So here's one of the simpler ones from figure one. This is the first figure in the book. And we can think about how like the active states, even though they're coming up from the right here, and they're coming kind of down from the left here. In both situations, it's like this is the engagement between the agent and the niche. Okay, but then there are some other representations too, like figure two, two. Here we see a proto blanket forming, though not described as such, and without like, for example, an arrow connecting these two nodes between observation and action. And more of an emphasis on the way that hidden states are being either actually generative of the observations, that's the generative process, or that hidden states of the generative model are doing inference on the observations. So that's one other representation. And then there's this representation. Where here action is coming out of discrepancy, as is perception, learning, generative model change, and we see observation coming into discrepancy. So, what is the action perception cycle? And where do Markov blankets come into play, in terms of even the classical ways of thinking about control systems? If one looks up control loop, they'll find all kinds of diagrams from cybernetics and control theory that can entirely be interpreted within a sort of entity, getting input, doing some sort of processing, and having outputs. So, how is the Markov blanket helping us approach this nexus of the entity in the niche? So, what is the case? So, what is the case? Well, one thought, or important piece to note, is that some blanket states will exist for any Bayesian graph that isn't fully connected. Anytime that you have a non-fully connected Bayes graph, so not every node is connected to every node, then there will be some other nodes upon which two other sets are conditionally independent. That's just what the nodes and edges mean in this kind of a graphical model, which is if nodes are random variables and edges are statistical relationships, then nodes directly

connected by an edge have a direct statistical relationship. Again, not implying anything about the material or the causal nature of the world.

So then, if two nodes do not have a statistical relationship, they're either part of two different graphs, which can be seen as basically partitioned from each other, or they're going to be part of a connected component of a graph, or they're going to be part of a graph. But there's some other nodes, one or more, that intermediate, such that those two nodes of interest, which are arbitrarily called internal and external, have conditional independencies. That ends up simplifying and being important for a lot of the computation, and it respects and reflects the sparse coupling of features in the world.

So it's always just important to think back, like, where is something new coming in to active inference? People have been using partially observable Markov decision processes for many years for perception and control issues before active inference.

So, those models also are imbued with Markov blankets, just like any other Markovian setting in time series analysis or any sort of partially observable model.

So, just an important question for discourse to continue on.

How is the Markov blanket restating the classical action perception cycle? Oli?

Just as a side note, I wanted to point out again to, I think, an important error in this particular figure, which I've included in the errata. So, if I could, and added a note there.

Yes.

So, if you could please open the errata for this. Yeah. And open the err notes there. I've added the explanation for why exactly. Yes.

Thank you.

So, let's just put this out.

It makes sense.

Yes. What? Giuseppe?

Well, it makes sense. But I think it's well spotted.

Yes. There's many changes. And so, again, these are like why it's so important for everybody to make asynchronous contributions and questions if they can. This is how we actually are going to move the needle on the textbook making sense and making impact. I know it's like an individual sense-making journey as well, but also these kinds of questions like, what does this mean? And the discourse around them is going to catalyze the actual use of this area.

Do you want to summarize what the difference is between the existing equations and the replacement?

Sure. You see, in the existing equation in the book, the marginal flow of sensory states are expressed in terms of blanket states, namely  $U$  and  $Y$ , and also the external state. But it should be the other way around. I mean, the marginal flow of sensory states should be expressed only in terms of external states and the blanket states. And the marginal flow of active states should be expressed in terms of only internal states and the blanket states. Thank you. So, yeah.

Just to restate that and also to build some intuition with what these flows are and how they relate. So,

U and Y constitute B, the blanket states. So, sometimes U and Y are both written, or B is written to describe the blanket state variables. Other times, it might just be describing a subset, at which point not the whole blanket is being described. Just, for example, the active states U or the sensory states Y are being described. Then,  $\mu$  are internal states here, and X, you know, find X, hidden states, external states, with a note here that the variables are different in different presentations. So, as it was written, we can read that this is the change in U, which is the active states, is a function of first a stochastic parameter about U itself, but then the deterministic component here is a function of the blanket states and the hidden states. So, it's a function of the flow function of the blanket states. And, the change in observations through time is, after accounting for a stochastic term, a flow function on blanket states and internal states. So, it's actually exactly the opposite with X and  $\mu$  being switched between these two. Because actions should reflect a flow over blanket and internal states, states, sometimes called the particular states, because that's the particle that's moving around. That's the Bayesian mechanical particle. And then, observations should reflect a flow on blanket states themselves and hidden states of the world. So, what does figure 3.1 mean? I think that's what we're looking at. How is it different than figure 2.5? And how is the agent's niche relationship described qualitatively and formally in active inference and other areas? So, I think we kind of explored that. People can add any more notes, but this is like a really important level to pull back to. Whether it's somebody who's having a question or even a well-intended or not criticism of active inference or a control theory, it can be useful. Are they describing challenges of the world? Like the map territory difference, like the complexity of the world always transcending the structure and complexity of our own models, or referring to the fundamental issues of taking action amidst uncertainty. Are people pointing to those issues and then part of the community describing the agent niche qualitatively and or formally? Or are they sort of suggesting or implying some kind of like hidden other way of thinking about how agents engage in the niche? Because there's a lot of questions that can arise. But these framings, although here the Markov blanket gives that kind of a Bayesian statistical and graphical stance on it and implementation opportunity space, which includes an incredible number of computational affordances like message passing, variational inference, and so on. This partitioning of the entity from the niche, figure from ground, is not diverging from the traditions of cybernetics and other areas. So just kind of like interesting, are people pointing to fundamental issues of perception and action, which in some senses is our field, and are they part of the building and the sense making around those scientific and meta-scientific models? Or is it just a position of like, well, it's not possible to model this? So, yeah, it's not possible to model this. Kind of random thought, but... Okay. Thank you. Yes, Ali.



Sorry, just one other brief side note about the term you used there, the particle particular states, which you mentioned is the joint set of the blanket states and the internal states.

The other related term used in the literature is the autonomous state, which refers to the joint set of internal and active states. Instead of the blanket states, we only take the internal and active states as the autonomous states or paths. That's just...

Yes.

I wanted to check that out as well.

Thank you.

This would kind of be a Fristen-esque phrasing, that the idea that these different sets are going to inherit from the particular partition.

So, the first move is the Bayes graph to the particular partition, which is just this specific partition.

If we're interested in the particular partition, if we're interested in the particular partition, then sometimes it's useful to talk about specific subsets of the particular partition.

If we were only interested in one of these categories, we would just call it by name, like talking about just external states or external state 11, dimension 11.

But other times we want to talk about partitioned states, but still refer to more than one type of state.

And so, Ali brought in this second clustering. The particular states, that's the particle that's moving around. That's the whole entity, internal states and blanket states. And then also times, it's important to consider the action and the internal states, because those are kind of the ones that can be thought of as

in a total control situation, they're the ones that are under direct control.

So, like you can control your mind and you can control your action.

Those are the autonomous states. Whereas you can't directly influence sensory states, but you might be able to take actions that modify sensory states by changing where you look or by changing some aspect of the world. So, those are autonomous states. And it relates to this question, can surprise only be diminished through action?

## What does anyone think?

[illegible]

Yep, this is the changed beliefs, which is speaking more broadly than personally held convictions.

This is going to be pointing towards the continuum between perception, learning, and deeper belief states in generative models.

So, perception is changing beliefs, Bayesian beliefs, which is to say, updating the generative model, learning and perception.

On the other hand, there's action, which can change the world.

And so, out of this discrepancy, it can be dissipated two ways.

It's kind of like if there was like a  $\Delta G$ , a Gibbs free energy, like something that was combustible, there's like a hill to go down energetically.

And so, now we're in an informational space, but there's still, at the clash of predictions and observations, is a non-zero discrepancy.

And from that discrepancy, there can be changes in the generative model parameters, reflecting beliefs, and also recourse to action.

And active inference is about how generative models, which include beliefs about action, resolve this kind of nexus through the flows on updating the model in action.

Go for blue.

Yep, go for blue.

Oh, sorry.

Yeah.

What is the difference between belief updating and mental action?

I'll add it as a question, and then someone can give a thought.

Any thoughts?

Well, maybe mental action refers to the active states of a more nested Markov blanket.

So, it's an action, it's a modulation of precision within the generative model, and it's not a motor action, you know, through motor effectors.

So, the blanket here is not made of the active and sensory states that actually communicate with the external world, but it's still internal.

That's, I don't know, that's my take on it.

Maybe wrong.

Okay.

Yeah, very nice.

I think just more discussion can be added.

So, just giving a quick thought.

Belief updating could refer to any kind of parametric updating at all.

It doesn't even have to be like a cognitive model.

Belief updating is just any time that there's a prior that meets with an observation and gets updated into a posterior, you have belief updating.

And we can use Bayesian surprise to talk about belief updating and so on.

And then, as pointed out, mental action, as it's used in the hierarchical models of it, is referring to the policy selection process that's hinted in the action,

but the policy selection process or module or archetype as oriented around mental variables.

So, it is almost out of necessity requiring a nested or a hierarchical model because there has to be like a sort of base layer that is not mental action.

But then, at the second level or above, you can start to see the same kind of pie policy variables and the

same way of thinking about affordances and free energy and expected free energy minimization processes in the second level or above,  
where those action-like processes or maybe even genuine action processes are referring to cognitive states rather than a motor effector,  
as would be understood in the case of like conventional motor action.

Whereas belief updating does not require a hierarchical model, it's just referring very, very broadly to Bayesian parametric change.

That makes sense. Thank you.

All right. So, can surprise be diminished through action?

No? Not only.

Like, you're surprised, you know, you expect it to be 20 degrees and you're observing it's 25.

You can either engage in action selection to change the thermometer under beliefs about how one's actions change the world, B matrix,

or one can just update their model to, that is now what I expect.

And, of course, that's the fun discussion in Active Inference with expect and prefer.

Because if organisms are in the game of diminishing or bounding surprise,  
then there has to be this nuanced dance between what is being observed,  
understanding it to be simply the most likely,

but you don't want to be on the likely train off the cliff.

So, there has to also be a space for slightly or very stubborn prior preference distributions  
that enable increasingly intense policies to be implemented

to actually shape the state of the world and oneself back towards that earlier expectation and preference  
rather than engaging in just purely cognitive updating.

That's just speaking about it generally, but there's a lot to unpack in a more model-specific way.

Okay, I'll leave shared a link.

Oh, awesome.

Awesome.

This looks like a good reference.

And also, Livestream 25, I believe, was on Mental Action.

Sanved Smith.

Okay.

Okay.

Let's look towards this question.

Does anyone want to just give an example that can clarify synchrony between internal and external states?

What's the situation where we can think of internal and external states synchronizing?

Okay.

Well, I guess it depends on what we mean by synchrony.

I mean synchrony like on a fast timescale or on a specific timescale or...

Because the fact that the internal states reflect the causal dynamics of the environment does not imply that they are immediately synchronized, right?

Yeah, you're absolutely right.

The timescale really matters.

And also, it's important to note that synchrony doesn't mean lockstep.

It doesn't mean that there's like a thermometer inside and a thermometer outside and they're just following each other.

Synchrony can refer to the kind of coordination that's more like orchestrated in the sense that if you had two people sawing a tree, they're both going to be taking opposite actions.

But there still is high mutual information between those two action policies.

And they're able to be under general synchrony.

So we can just think of a few areas of examples.

Like one can always think of a computer example because computers are clean.

Like you could just say, well, I have two servers and they're communicating on the time.

Or they're sending each other information back and forth.

Or they're tracking some feature about each other.

You could imagine a communication example where there's some kind of semantic coordination or synchrony.

You could also imagine some kind of a cellular or a physiological synchrony.

Like the inside of the cell is being modeled as doing inference about insulin levels outside of the cell.

And then the insulin receptor is like the interface.

And so there's synchronization of cellular response to perceived or inferred insulin concentration.

And the external unobserved true circulating insulin concentration mediated through set states, the binding kinetics of insulin on its receptors.

Like these are just kind of qualitative examples, but there's a lot to drill down into more.

Ali?

Ali?

Ali?

I think the word synchrony here might have been used in the technical term of the existence of a kind of synchrony map

between the two coupling systems.

So in order for two systems to be coupled, there needs to be a synchrony map, namely sigma between them.

And the whole paperwork example, paper by Dalton Sack-Divariwell, provides multiple physical examples for these kind of systems, coupled systems.

And the most extreme case of those coupled systems are the sparse coupled systems, in which the Markov blanket is the strongest, or as they call it,

the blanketedness or the blanket index is basically zero.

But in some other kinds of systems, which in which the internal and external states are somehow more closely follow each other, the Markov blanket tends to weaken and diminish.

So, yeah, I think that's the meaning of the word synchrony here.

Great point.

Yep, livestream 49.1 with Jakob and Ali this week.

It's a recommended listen.

And it's quite far-ranging.

So that is the most state-of-the-art addressment of the issue.

A slightly older addressment of this issue.

And also, importantly, coming back to what Ali mentioned about the synchronization manifold.

So here's the example of Birdsong that Fristen and Fristen and Frith have worked on.

And this is some of the colorized versions of the trace diagrams that are seen in the textbook that are like a little bit difficult to tell what's happening when they're uncolored lines.

But just to highlight the manifold there, the x-axis and the y-axis are the first and the second bird's belief states.

And so to the extent that those pairs of beliefs at a given moment exist on the  $y = x$  line, they're believing the same thing at the same time.

Whereas if you could imagine if they were believing opposite things at the same time, you'd have like a negative regression.

And if there was like uncorrelated or unsynchronized belief updating, you'd have just a scatter cloud that wouldn't have a significant regression slope.

And so this diagram is showing how before and after learning, the sharpness of the manifold increases.

And a manifold is a lower dimensional space within a higher dimensional space.

So like we have a two-dimensional space here with  $x$  and  $y$ .

And then there's activity only on this manifold, which is being projected effectively, not completely, down into one dimension, because it's like a number line.

Or you can imagine it was a three-dimensional space that was being projected into a two-dimensional manifold.

So manifold is just some lower dimensional structure in a higher dimensional phase space.

And so synchrony, as Ollie mentioned, is referring quite particularly technically to the existence of synchronization manifolds.

And that is related to sparse coupling, blanket index, all these topics.

All right.

So in our last couple minutes, we'll look at this question about 3.4.1.

And then over the coming week, just chill, enjoy the week.

Also add a few other questions or discourse so that we can come back to some of the latter sections of chapter 3 next week.

So let's look at 3.4.1.

In 3.4.1, that's the section, variational free energy, model evidence, and surprise, the authors look at the equivalence between variational free energy, model evidence, and surprise, and formalize this relationship in equation 3.2.

Is it possible to unpack equation 3.2 to make these relationships easy to understand?

Great question.

All right.

So let's go to equation 3.2.

Here's what the equation looks like.

Then,

our

extremely helpful

annotation crew

has gone through and provided this natural language

unpacking.

So I'm just going to copy it down, and then we can look at what it actually says,

and then see how that relates to the question that was asked,

which was about unpacking the relationship.

Okay.

So the surprise fracture I, fancy I, of the data Y conditioned vertical line on the model M

is equal to

the negative

natural log LN

of the conditional probability,

that's P,

of the data Y given M

with its upper bound being,

so

this is a less than or equal to sign,

so they could be equal,

these two, the first and the second one,

are equal.

The second one and the third one

could be equal,

or

the third term could be larger.

Its upper bound

being the KL divergence,

DKL,

between

two lines,

the approximate posterior of hidden states

$Q$  of  $X$ ,

and

the conditional probability  $P$

of

hidden states  $X$

conditioned

vertical line

on the data and the model

minus

the natural log

of the model evidence

$P$  of the  $Y$

given the model.

Evidence for the data

given the model.

Okay.

So the question was,

what are the relationships

between variational free energy,

that's the third term,

model evidence,

the second term,

and surprise,

the first term?

the answer is not in the simple reading

of the equation,

but

that

gets us to

the discussion point.

So how are these terms related?

The three labeled terms on the bottom.

A related question is,

why do we use variational free energy?

What is effective or useful

about energy measures at all?

Okay.

I'll give one thought

and then one or two other people

could add in,

which is

we

don't

want to be

outside of our range

of surprise.

And so

by defining surprise

in terms of data

and specifically

within the natural log

space of data,

which has some

just nice properties,

we

want to stay bounded

in the model evidence.

Like we want to have high

or the highest

or at least adequate

model evidence.

And that is equivalent

to surprise.

We can see the same

Y

given M.

Now,



if we had that  
and had confidence  
without a doubt  
in our model  
and we knew how to calculate  
the likelihood function  
of the data,  
then  
things would be easy.  
And there are situations  
where things are easy like that.  
But there's other cases  
where  
we can  
bound  
our surprisal,  
which is to say  
bound  
our model evidence,  
evidence lower bound,  
the ELBO,  
E-L-B-O,  
and we can bound  
this intractable quantity  
model evidence  
with  
a divergence-based  
energy  
calculation.  
And so we can do  
gradient descent,  
variational inference,  
message passing,  
active block inference,  
all these amazing things  
on the variational  
free energy  
and just know

that at best  
we're going to converge  
to the model evidence.  
So this is  
an optimization,  
an optimizable  
quantity.  
of variational  
free energy.  
Whereas model evidence  
is like,  
if you know it,  
you know it.  
But if you don't,  
you're kind of stuck.  
Like,  
how likely  
is this book?  
And it's super hard  
to understand  
how changing  
one word  
would change  
the overall  
likelihood distribution.  
And so  
that motivates  
approximate  
heuristics  
for Bayesian inference,  
especially those  
that are authorized  
by the sparsity  
of the data.  
And the variational  
free energy measure  
allows us  
to bound

surprise  
with attractable,  
optimizable  
function  
rather than  
needing to,  
for example,  
flesh out  
the total space  
of data likelihoods  
and model likelihoods  
which might be  
futile.

I'm sure  
there are other  
better ways  
to add things.

So,  
one thing  
that,  
like,  
after reading it,  
after reading about,  
after reading a book  
and after reading  
other article,  
I kind of had  
an insight,  
but I don't know  
if it's correct.

So that  
the whole energy  
stuff,  
of course,  
comes from  
the physics.

And  
it's very

legitimate  
to be  
a bit,  
a bit,  
a bit confused  
because  
this formulation  
of,  
with the  
Kulbeck-Liber  
divergence  
is  
a purely  
statistical  
quantity.  
So,  
it's  
understandable  
that somebody  
says,  
what does  
energy  
have to do  
with that?  
It seems  
to me  
that  
the  
two,  
the  
high road  
and the  
low road  
converge  
here  
because  
when you  
minimize,

on one  
hand,  
you can  
see,  
you can  
say that  
the  
organism  
maintains  
its forms  
and function  
by minimizing  
surprise,  
and that  
is  
maximizing  
model evidence.

But on the  
other hand,  
if you do  
this,  
you're going  
to be able  
to compute  
the approximate  
posterior,  
a good  
approximate  
posterior  
about the  
states,  
the  $Q$   
and  $X$ ,  
the  $Q$   
of  $X$ .

So,  
in a sense  
here,

surprise  
minimization  
is also  
a way  
to compute  
a good  
posterior  
over the  
states.  
all right,  
great,  
thanks for  
that.  
No,  
we'll  
return to  
this.  
Yes,  
let's  
return to  
it,  
pick up  
here,  
and any  
other  
questions  
in next  
week.  
So,  
thank you.  
See you  
all later.  
Thank you.